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The Effect of Core and Carburized Surface Microstructural Stability on Residual Stress Evolution during Tempering

ABSTRACT: It is a well-known fact that compressive residual stresses are developed during the quenching process of carburized parts. The main purpose of this work is related to the tempering effects on as-quenched compressive residual stresses of commercial carburized samples. The work was carried out by using SAE 4120 RH steel samples with different diameters (15 and 38 mm), which led to different cooling rates with a direct effect on the nucleus microstructure. The carburized surface microstructure was considered similar. The 15 mm bars nucleus microstructures were predominantly martensitic but bainitic in the larger samples. The as-quenched surface compressive residual stress was lower in the smaller diameter samples. The compressive residual stress reduction due to increasing tempering temperature was continuous for the larger diameter work piece. The smaller diameter samples had different behaviors in two regions; the compressive stress for lower tempering temperatures was reduced, while values were enhanced over higher temperatures.

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Introduction

Residual stresses can be analyzed from macroscopic and microscopic standpoints. Metallurgical defects like dislocations or even point defects promote localized residual stresses that can be defined as short range stresses. Even in the microscopic standpoint, dislocation can develop long range stresses like the pileup back stress effect, which is still a small distance interaction, if compared to the macroscopic stresses that can be developed by plastic deformation, temperature gradients, or phase transformation like martensitic reactions [1,2].

The macroscopic residual stresses can be produced by plastic deformation gradients due to mechanical or thermal loading, as well as phase transformation occurring in different component parts. The quenching process promotes macroscopic residual stresses due to thermal and phase transformation effects, as well as short range of phase transformation internal stresses. In some cases, the magnitude of the macroscopic residual stresses can overlap the material tension rupture resistance inducing fracture processes, for instance, quenching cracks [2].

The relationship between the M_s temperature and the steel chemical composition has been extensively studied. Equations 1–3 (Refs 3–5, respectively) can be used as examples; however, the main chemical element in terms of lowering the M_s temperature is carbon

$$M_s = 539 - 432(\% \text{ C}) - 30,4(\% \text{ Mn}) - 17,7(\% \text{ Ni}) - 12,1(\% \text{ Cr}) - 7,5(\% \text{ Mo}) \quad (1)$$

$$M_s = 520 - 320(\% \text{ C}) - 50(\% \text{ Mn}) - 30(\% \text{ Cr}) - 20[(\% \text{ Ni} + \% \text{ Mo})] - 5,0[(\% \text{ Cu} + \% \text{ Si})] \quad (2)$$

$$M_s = 561 - 474(\% \text{ C}) - 33(\% \text{ Mn}) - 17(\% \text{ Ni}) - 17(\% \text{ Cr}) - 21(\% \text{ Mo}) \quad (3)$$

The effect of the thermochemical heat treatments on low carbon steel residual stress developed in components has been well-known for many years mainly in the automotive industry, in parts like roller bearings and gears. Normally the stress developed in the surface is compressive and enhances the fatigue resistance of the mechanical parts. The compressive state of stress developed during quenching is generally related to the higher carbon content at the surface which lowers the M_s temperature promoting a surface martensitic transformation after the nucleus, as shown in Fig. 1.

As it can be seen in Fig. 1, the last martensitic reaction occurs at the surface, and its expansion restriction due to the previously reacted nucleus produces compressive tangential stresses in the surface.

The real tension distribution during a component loading is composed by the project stresses plus the residual ones [6], as seen in Fig. 2.

In the automotive power transmission industry, the thermochemical heat treatment applied to the gears manufacture is the carburizing process. The compressive residual state of stress achieved by this process enhances the fatigue resistance of these components [7], but the long term temperature exposure during vehicles in operation can affect the compressive residual stress.

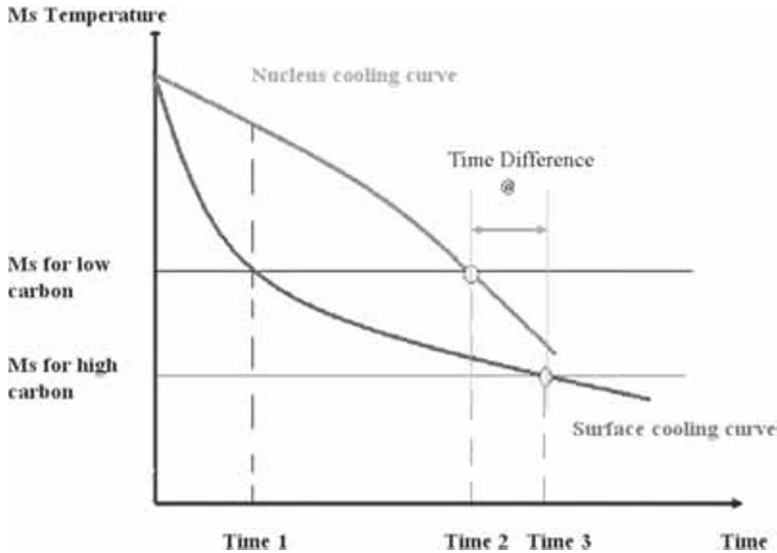


FIG. 1—Surface and nucleus cooling curves related to the M_s temperatures for low and high carbon contents.

Compressive residual stresses are normally enhanced in gear production by the shot peening process, but this will not be taken into account in the present study, which is focused specifically on stresses developed by heat treatment.

The working temperature of vehicle transmission gears lies between 90 and 110°C, but the vehicle's long useful life can be responsible for stress relief and solid state reactions that are usual in short time exposure to higher temperatures. Cyclic loading can accelerate the residual stresses fading even at room temperature cyclic test run on a shot peened surface of AISI 5160 steel quenched and tempered to 45 HRC [8].

In the present work, the temperature effect was studied by isochronal heat treatment, being the temperature only the variable from 180 to 500°C. The engine oil temperature lies below the tempering temperatures used in this study, but as previously mentioned, the long term running (with cyclic loading in real systems) may produce stress relief and solid state reactions that occur in short time higher temperature treatments.

The important issue in the present study is the discussion regarding the residual stress evolution during tempering, taking into account the global solid state reactions taking place in the case carburized and nucleus low carbon steel.

Experimental Technique

The samples subjected to the thermochemical process are shown in Fig. 3.

There are 20 total samples, 10–15 mm bars, and 10–38 mm bars. The main reason for using different diameter bars is to introduce in this study a size

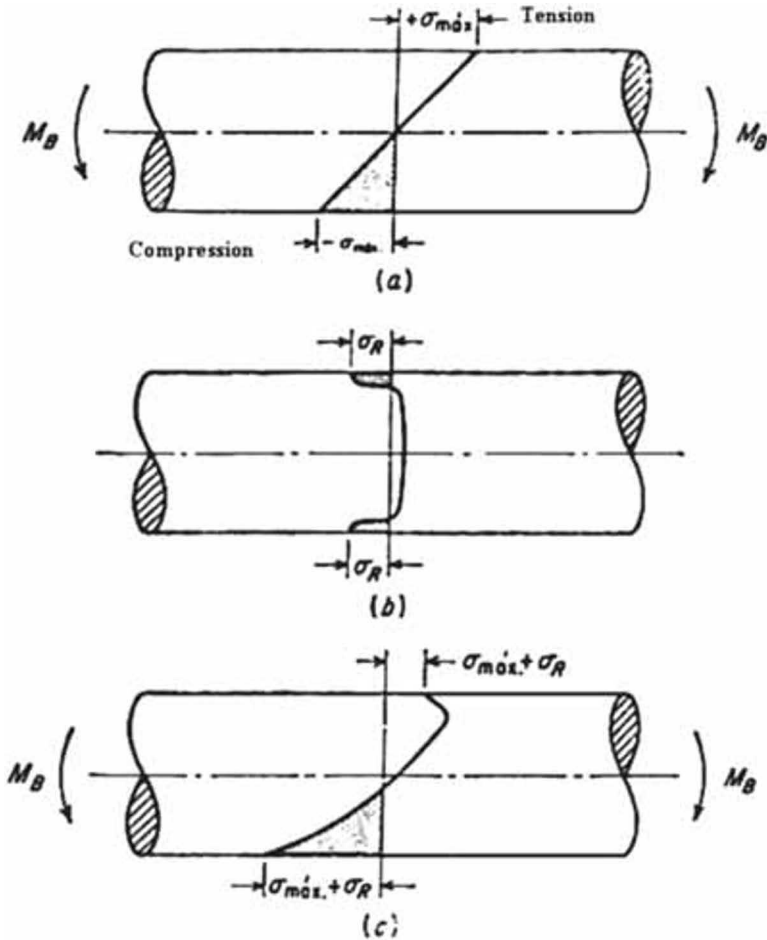


FIG. 2—Stresses developed during flexural loading of a bar that has compressive residual stresses [6].

effect on the residual stress behavior related to different gear sizes.

The material composition of the bars can be seen in Table 1.

The steel chemical compositions are related to the SAE 4120RH with a small chemical adjustment imposed by the transmission manufacturer.

The carburizing treatment was performed in a chamber furnace during 6 h at 930°C with a 1.2 % of carbon potential. The next step was a quenching process at 845°C, using as cooling media an oil at 95°C.

After the quenching process, longitudinal residual stresses were measured in all samples by an x-ray diffractometer: the Digital FASTress-Residual Stress Analyzer. The x-ray source was Cr-K α , with an incidental angle ranging from 0 to 45° and an irradiated area from 1 to 4 mm. For each bar diameter, three residual stress measurements were made for each tempering sample. There

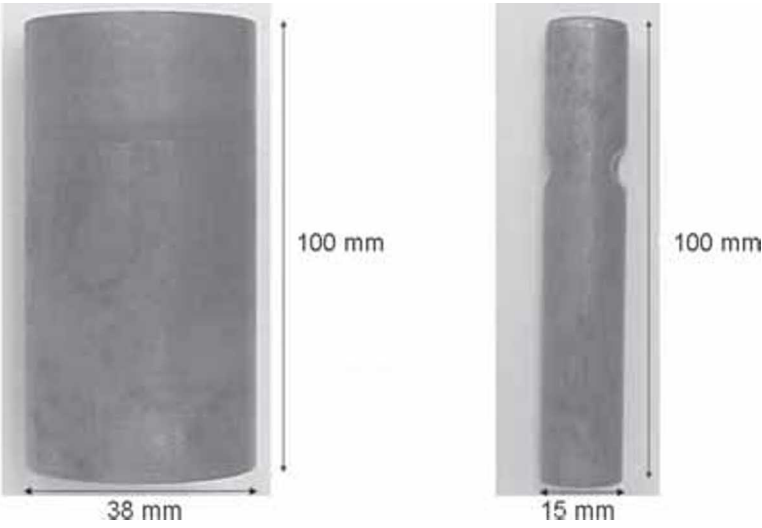


FIG. 3—At the right side a 15 mm diameter bar; at the left, a 38 mm diameter bar.

were two samples for each tempering temperature, starting from 180 to 500°C within a range of 80°C. The residual stress for the quenched conditions was determined by ten measurements, one measurement per each sample.

Rockwell C Hardness tests were performed on the surface and core after quenching and tempering treatments in three different areas of each sample. One specimen was prepared in order to measure the core hardness for each temperature.

Metallographic samples were prepared for all heat treatments conditions, but only the quenched one will be presented in this study.

Results and Discussion

Table 2 is related to the surface hardness results for the 38 and 15 mm carburized bars in the as-quenched and tempered conditions.

The results presented in Table 2 show that the surface hardness conditions are similar for the two bars of different diameters. The increase in the tempering temperature reduces the surface hardness in the same way as in the 15 and 38 mm diameter bars. This behavior indicates that the carburized microstructure can be considered the same for the two bars under study. The microstructure of the case hardened surfaces for the two bars can be seen in Fig. 4.

TABLE 1—Chemical composition (wt %).

C	Mn	Cr	Mo	P	S	Si
0.21	1.16	0.49	0.13	0.02	0.024	0.18

TABLE 2—*Surface hardness.*

Temperature (°C)	15 mm Diameter		38 mm Diameter	
	Average of Surface Hardness (HRC)	Standard Deviation	Average of Surface Hardness (HRC)	Standard Deviation
As-quenched	64.9	0.1	64.6	0.3
180	62.2	0.1	62.8	0.1
260	58.5	0.1	59.7	0.1
340	53.0	0.6	53.2	0.6
420	48.5	0.6	49.6	0.6
500	42.8	0.6	42.7	0.6

Instead of x-ray retained austenite determination, the estimated content for the gears manufactured was about 5 %.

The core hardness results for the 38 and 15 mm carburized bars in the quenched and tempered conditions can be checked in Table 3.

Table 3 shows that the core as-quenched hardness of the lower diameter bar is about 15 HRC higher than the 38 mm diameter core bar. The tempering temperature reduces significantly the core hardness in the 15 mm bar, having a slight effect on the 38 mm specimen. This different behavior shows that the larger diameter bar core microstructure is stable as related to the lower diam-

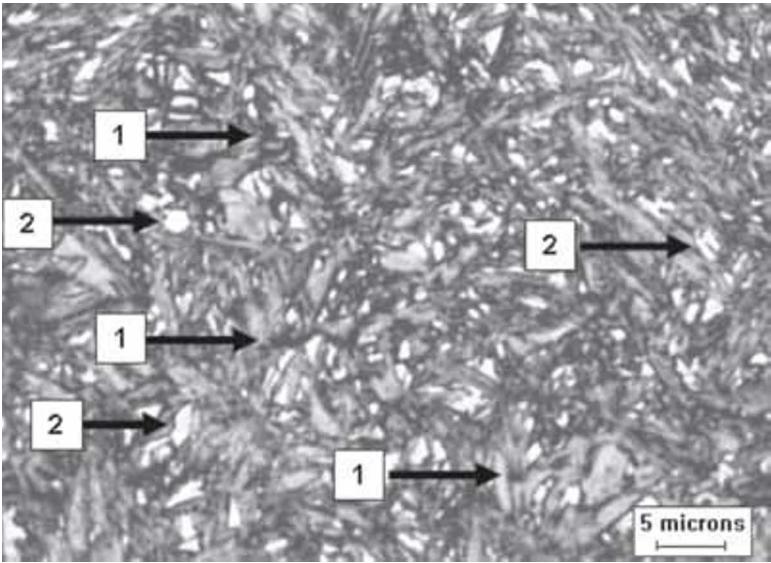


FIG. 4—*Carburized quenched surface microstructure for 15 and 38 mm bars. Labeled arrows (1) represent tempered martensite plus thin carbides. Labeled arrows (2) illustrate small retained austenite quantities.*

TABLE 3—Core hardness.

Temperature (°C)	15 mm Diameter		38 mm Diameter	
	Average of Core Hardness (HRC)	Standard Deviation	Average of Core Hardness (HRC)	Standard Deviation
As-quenched	37.1	0.2	22.1	0.2
180	35.7	0.4	21.9	0.6
260	34.5	0.5	21.4	0.1
340	34.2	0.5	21.1	0.3
420	32.4	0.8	21.0	0.1
500	26.8	0.6	19.3	0.7

eter ones. Figures 5 and 6 show the core microstructure for the 15 and 38 mm diameter bars in the as-quenched condition.

The core microstructures for the 15 and 38 mm bars are different, being mainly bainitic (upper bainite) for the larger specimen. The martensitic smaller diameter bar structure can be mixed with some bainite (lower bainite); however, their behavior in terms of solid state reactions during tempering can be considered similar. The core upper bainite structure in the large bars will be the

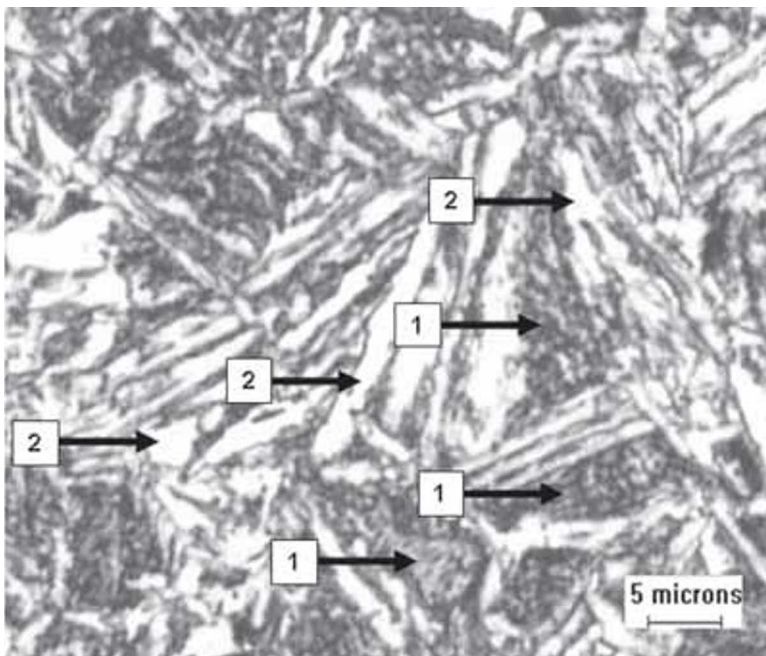


FIG. 5—Core microstructure of the 15 mm bar. Labeled arrows (2) show the upper bainite structure. Labeled arrows (1) illustrate the martensitic structure.

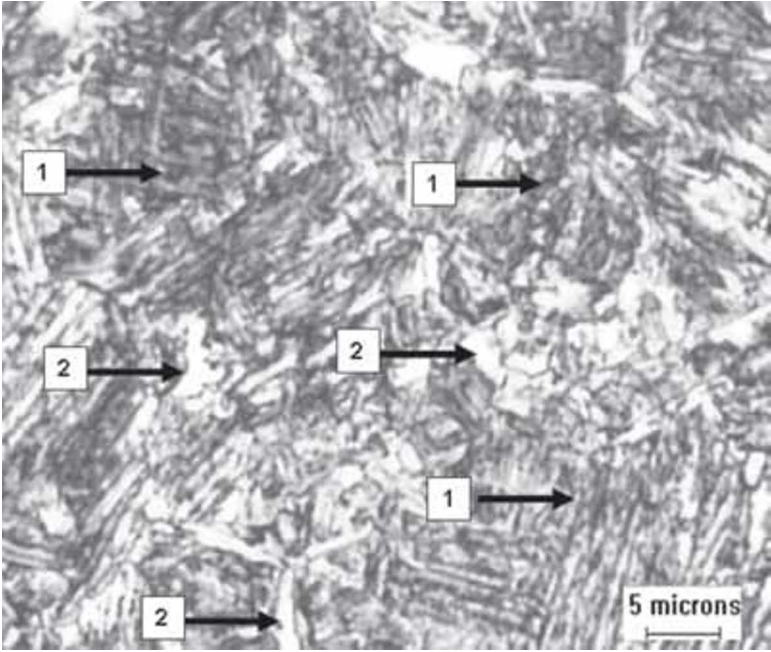


FIG. 6—Core microstructure of the 38 mm bar. Labeled arrows (1) show the martensitic structure. Labeled arrows (2) show small quantity of proeutectoid ferrite.

basis for the differences observed in terms of residual stress evolution during tempering in the higher temperature range, as it will be discussed later.

Table 4 is related to the compressive residual stress results for the 38 and 15 mm carburized bars in the quenched and tempered conditions.

This discussion will be divided in two parts.

TABLE 4—Modulus of tangential surface compressive stress in MPa.

Temperature (°C)	15 mm Diameter		38 mm Diameter	
	Compressive Residual Stress Result in Module (MPa)	Standard Deviation	Compressive Residual Stress Result in Module (MPa)	Standard Deviation
As-quenched	156.7	13.3	325.4	3.0
180	145.2	18.2	305.4	35.6
260	66.6	4.1	203.4	8.5
340	100.8	34.3	162.5	8.0
420	134.0	17.7	130.7	17.2
500	118.0	18.3	135.7	51.8

Explanation of the Quenching Stress

For the same cooling conditions, the time difference for martensitic reaction between nucleus and surface (@ in Fig. 1) diminishes considering larger cross sections. This effect can reduce the surface compressive residual stresses that are normally developed when the surface martensitic reactions are effectively the last phase transformation in fully martensitic cross sections.

Table 4 result shows that in these experiments, the larger diameter samples produce higher compressive residual stresses as compared to the smaller diameter ones. A good explanation for this effect lies on the core microstructure, mainly martensitic in the 15 mm samples and predominantly upper bainite for the 38 mm bars. As mentioned in the introduction, upper bainite can be formed in a temperature range between 600 and 650°C, higher than the low carbon martensite reaction. Thus, the core upper bainite reaction can occur before the martensitic surface transformation due to the higher temperature reaction, increasing the compressive residual stresses. The core martensite volume fraction was small in the 38 mm bar and was not sufficient to compensate for the nucleus contraction during the martensitic transformation, whose global effect enhances the compressive tangential stresses in the surface.

Evolution of the Residual Stresses during Tempering

To develop this discussion we will assume that stress relieving processes in case stresses will take place in the same way as in the two bars due to the similarity between their structures. For the low temperature tempering range (180–340°C), the short term hot hardness of a AISI 52100 quenched and tempered steel is above 46 HRC [9], which means a high yield stress. Assuming a similar behavior for the case hardened surface, the compressive residual stress will be many times smaller than its yield stress, reducing the effectiveness of the stress relieving process. For this reason, the effectiveness of the solid state reactions taking place in the surface and core will be considered as an important factor to explain the different behavior between the two bars, mainly considering that the difference between these specimens is related to the core microstructure, as showed in Figs. 5 and 6.

The tempering reactions on the ferrous matrix volume can be summarized as follows.

The epsilon carbide precipitation (0–200°C) as well as the cementite formation directly or due to the epsilon carbide decomposition (230–350°C) reduces the specimen size.

The retained austenite decomposition (200–300°C) results in an increment in the specimen size.

The expansion and contraction effects depend on the relative influence of each reaction as well as on the ferrous matrix volume fraction and amount of carbides involved in each solid state reaction [1,10].

Figure 7 presents the microstructure and residual stress evolution in a graphical way to improve the result discussion.

*Region 1 (Low Tempering Temperature Range): 15 and 38 mm with Similar Reducing Stress Behavior—*The 38 mm bar core microstructure is stable (upper bainite); the 15 mm bar with core microstructure is close to stability, no epsilon

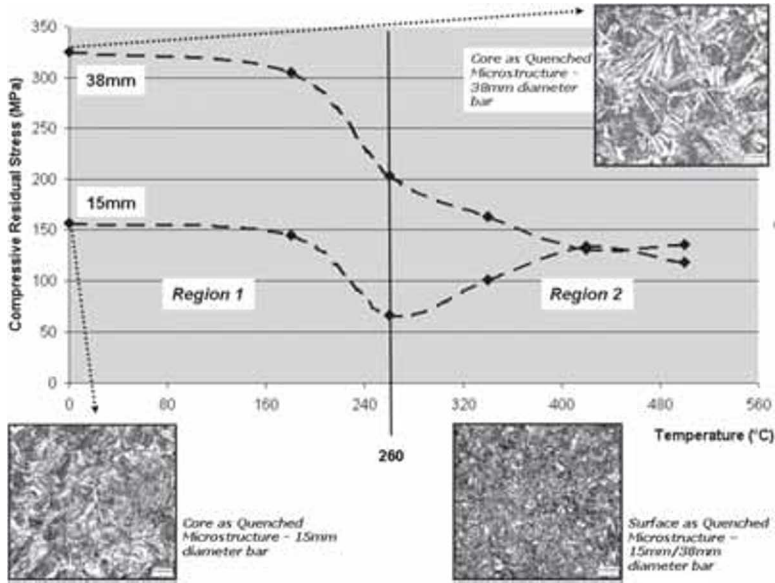


FIG. 7—The residual stress evolution indicates two distinct regions named Region 1 and Region 2.

carbide precipitation due to the low carbon content ($\sim 0.2\%$), cementite precipitation at the beginning, and absence of retained austenite. On the other hand, in the carburized structure the high carbon content prompts the epsilon carbide precipitation and the start of its decomposition in cementite, leading to a contraction contributing to the lowering of the compressive stresses. The austenite expansion effect can be neglected due to its low content in this structure.

Region 2—This part of the discussion is strongly related to the core structure because the case carburized structure is similar to the two work pieces, which means the same time temperature solid state and stress relieving behavior.

The 38 mm bar core is still stable due to the upper bainite structure and low martensite content. The lowering of the compressive stress modulus observed can be associated with the completion of the cementite precipitation and some stress relieving effect on the carburized surface.

The 15 mm bar tempering response is different with an increase in the compressive residual stress modulus, whose explanation takes into account the different core microstructure as related to the 38 mm bars, reducing its volume due to the cementite precipitation in the unstable martensitic structure.

Conclusions

The carburized surface microstructure was similar for the 15 and 38 mm bars, with no significant differences in terms of hardness in as-quenched and tempered conditions.

Core structure in the 38 mm bar was mainly bainitic (upper bainite). In the 15 mm bar the main core structure was low carbon martensite.

The as-quenched compressive residual stress modulus was higher for the 38 mm bars.

The compressive residual stresses were reduced after tempering the quenched bars.

The residual stresses reduction due to the tempering process was different between the two bars in the higher tempering temperatures due to the differences in their core microstructures, mainly martensitic in the 15 mm diameter samples and bainitic (upper bainite) for 38 mm bars.

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